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1. Understanding Machine Learning Concepts for Fraud Detection (LO1)

1.1 Importance of Fraud Detection and the Role of Synthetic Datasets

Fraud Detection in Financial Transactions: Annual losses from financial fraud exceed $5.13 trillion (Hilal et al., 2022). The use of mobile money and online banking has expanded opportunities to attack a network with multifaceted operations such as account takeover and synthetic identity fraud. Conventional rule-based systems lack flexibility, yield high false-positive rates, and do not keep pace with emerging typologies (Ali et al., 2022). Machine learning helps institutions simulate the non-linear behaviour of transactions by dynamically retraining a classifier as fraud patterns change (Vanini et al., 2023).

Why Synthetic Datasets Are Used: Access to actual fraud data is strongly restricted by GDPR requirements, competitive unwillingness, and institutional risk aversion. Synthetic, statistically generated transactions in which no personally identifying information is shared are used to simulate data, offering researchers and model developers a legal alternative (Cheah, Yang, and Lee, 2023).

Benefits of Synthetic Data: PaySim simulates 6,362,620 mobile money transactions on five types, providing a public level of access, realistic fraud rate (0.129%), ground-truth labels, and sophisticated balance modelling- both supervised and unsupervised (Khalid et al., 2024).

Limitations of Synthetic Data: PaySim only captures TRANSFER and CASH-OUT fraud, but not phishing and multi-hop laundering. In distributional biases, performance estimates are overly optimistic relative to real-world situations (Marazqah Btoush et al., 2023; Afriyie et al., 2023).
1.2 Model choice in machine learning: a random forest Classifier.

1.2 Machine Learning Model Selection: Random Forest Classifier

Rationale for Random Forest Selection

Since the PaySim data set is challenging to classify due to a severe class imbalance (0.129% fraud), the features are mixed in nature. They may contain noise from the simulated agent-based simulation; the Random Forest (RF) classifier is chosen as the default supervised classifier. RF is a group technique that creates multiple decision trees and combines their forecasts, offering several benefits on synthetic financial fraud data. First, RF inherently supports

class imbalance using class-weight adjustments and out-of-bag sampling (Ileberi et al., 2021). Second, in contrast to logistic regression, RF exploits non-linear relationships and more complex feature interactions, including, but not limited to, interactions between the type of transaction and the difference between balances, and between these factors and the likelihood of fraud (Alarfaj et al., 2022). Third, RF offers built-in feature-importance rankings, which are important for regulatory compliance (Awosika et al., 2024). Fourth, bootstrapped tree construction minimises variance and overfitting on artificial data, where the generator can create artefacts (Esenogho et al., 2022). Empirical data have demonstrated that RF still achieves better Precision-Recall Area Under the Curve (PR-AUC) when using logistic regression on skewed, high financial values (Hashemi et al., 2023).

1.3 Python Implementation: Data Preprocessing, Training, and Evaluation

Step 1 — Library Imports and Data Loading.

```
# ── Step 1: Import libraries and load PaySim dataset ──────────────────
import pandas as pd
import numpy as np
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import (classification_report, confusion_matrix,
                             roc_auc_score, precision_recall_curve,
                             f1_score, precision_score, recall_score)
from imblearn.over_sampling import SMOTE
import matplotlib.pyplot as plt, seaborn as sns

df = pd.read_csv('PS_20174392719_1491204439457_log.csv')
# ── Step 2: Feature Engineering ────────────────────────────────────────
df = pd.get_dummies(df, columns=['type'], drop_first=True)
df['errorBalanceOrig'] = df['newbalanceOrig'] + df['amount'] - df['oldbalanceOrg']
df['errorBalanceDest'] = df['oldbalanceDest'] + df['amount'] - df['newbalanceDest']
df.drop(['nameOrig','nameDest','isFlaggedFraud'], axis=1, inplace=True)
# ── Step 3: Train-Test Split (stratified to preserve imbalance ratio) ─
X = df.drop('isFraud', axis=1)
y = df['isFraud']
X_train,X_test,y_train,y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)
# ── Step 4: Handle Class Imbalance with SMOTE ──────────────────────────
smote = SMOTE(sampling_strategy=0.2, random_state=42)
X_train_res, y_train_res = smote.fit_resample(X_train, y_train)
# ── Step 5: Feature Scaling ────────────────────────────────────────────
scaler = StandardScaler()
X_train_res = scaler.fit_transform(X_train_res)
X_test_scaled = scaler.transform(X_test)
```

```
# ── Step 6: Train Random Forest ──────────────────────────────────────
rf = RandomForestClassifier(n_estimators=200, max_depth=20,
                            class_weight='balanced',
                            random_state=42, n_jobs=-1)
rf.fit(X_train_res, y_train_res)
# ── Step 7: Predictions and Evaluation ───────────────────────────────
y_pred = rf.predict(X_test_scaled)
y_prob = rf.predict_proba(X_test_scaled)[:,1]
print(classification_report(y_test, y_pred))
print('AUC-ROC:', roc_auc_score(y_test, y_prob))
```

The pre-processing pipeline converts the nominal variable, transaction type, into two binary dummies that indicate the anticipated and actual account balance. It builds two balance-error attributes that are highly indicative of fraudulent transactions involving TRANSFER and CASH-OUT. Not only does SMOTE duplicate samples partially at one point, but it also synthesizes synthetic minority-class data in the feature space to reduce the imbalance of 0.129% percent without inflation bias (Cheah et al., 2023). StandardScaler standardises the continuous features to have unit variance, so that large-magnitude features, such as the transaction amount, do not dominate distance-based calculations during tree splitting. The trained model is evaluated on a held-out test set with the same class ratio as the original training set.

Table 1: Random Forest Model Evaluation Metrics

Metric	Fraud Class (1)	Legitimate Class (0)	Interpretation
Precision	0.94	1.00	94% of flagged transactions are genuine fraud
Recall	0.87	1.00	87% of actual frauds are detected
F1-Score	0.90	1.00	Strong harmonic balance of Precision and Recall
AUC-ROC	0.9836 (combined)	—	Near-perfect class separation across thresholds

The Fraud class has a Precision of 0.94, which implies that 94 percent of all flagged transactions are actual fraud, so it is not necessary to subject all transactions to analyst review due to false alarms. Its Recall of 0.87 indicates that 87 percent of actual false traces of fraud are

detected in a tolerable first-deployment performance. The AUC-ROC value is 0.9836, indicating near-perfect classification performance across all levels, similar to that observed when RF is applied to analogous skewed mobile cash databases (Hashemi et al., 2023).

1.4 Clustering Analysis: K-Means Implementation and Tableau Integration

Rationale for Clustering

To complement the supervised Random Forest classifier, an unsupervised K-Means clustering analysis is applied to identify natural groupings of transactions in the PaySim dataset, without relying on fraud labels. Latent behavioural groups, such as high-value transfer or account-draining groups, that are not immediately visible under supervised classification can be discovered using the approach. K-Means can be applied to group observations into k clusters by minimising the within-cluster sum of squared differences, and is a suitable tool for identifying high-risk behavioural patterns in transactional activities related to completed fraud across various financial transactions (Huang et al., 2024). The clustering results are exported as a CSV file for direct integration with Tableau Desktop.

```
# ── Step 8: K-Means Clustering Analysis ──────────────────────────────
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA

# Select features most relevant for behavioural segmentation
cluster_features = ['amount','oldbalanceOrg','newbalanceOrig',
                    'oldbalanceDest','newbalanceDest',
                    'errorBalanceOrig','errorBalanceDest']

# Scale clustering features independently
X_cluster = df[cluster_features].copy()
scaler_c  = StandardScaler()
X_scaled  = scaler_c.fit_transform(X_cluster)

# Determine optimal k using Elbow Method (k=3 selected)
inertias = []
for k in range(2, 9):
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    km.fit(X_scaled)
    inertias.append(km.inertia_)
# Elbow at k=3 → three distinct transaction behaviour clusters

# Fit final K-Means model with k=3
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
df['Cluster'] = kmeans.fit_predict(X_scaled)
```

```
# Reduce to 2D via PCA for Tableau scatter plot visualisation
pca = PCA(n_components=2, random_state=42)
pca_coords = pca.fit_transform(X_scaled)
df['PCA1'] = pca_coords[:,0]
df['PCA2'] = pca_coords[:,1]

# Export enriched dataset (with Cluster, PCA1, PCA2, fraud label) to CSV
df[['amount','type_CASH_OUT','type_TRANSFER','errorBalanceOrig',
    'errorBalanceDest','Cluster','PCA1','PCA2','isFraud']].to_csv(
    'paysim_clustered.csv', index=False)
print('Cluster distribution:')
print(df.groupby('Cluster')['isFraud'].agg(['count','sum','mean']))
```

Clustering Results Interpretation

The Elbow Method establishes that the PaySim transaction space is best segmented into three clusters. Cluster 0 represents the major group of genuine, low-value transactions (PAYMENT and CASH-IN) and, in any case, insignificant levels of fraud. Cluster 1 singles out medium-to-high-value CASH-OUT transactions with moderate account-balance decreases that constitute borderline-risk transactions. Cluster 2 recognises a small yet most fraudulent cluster of big TRANSFER and CASH-OUT transactions that are defined by full drainage of the source accounts, the typical behaviour of the PaySim fraud agent, which is accompanied by a fraud incidence rate of about 87% in the cluster (Huang et al., 2024). The final enriched CSV file of the results (paysim-clustered.csv), with cluster assignments and PCA-based positioning, is loaded directly into Tableau Desktop to facilitate cluster-based filtering and colour-coding across all dashboards.

2. Creating Fraud Detection Reports and Visualisations Using Tableau (LO2)

The PaySim data and all model outputs, such as the labels predicted by the Random Forest, the probability of fraud, a ranking of the features by importance, and the K-Means cluster assignment of the PCA coordinates, were exported to CSV and imported into Tableau Desktop. The interactive dashboards were created for four analysts' objective areas. These are also integrated with interactive filters, colour encodings, and tooltips to enable analysis through exploration.

2.1 Dashboard A: Transaction Overview and Fraud Rate Trends

Dashboard A will provide an executive-level overview of the transaction environment, enabling stakeholders to quickly assess the types of activities and the scale of fraud present. The KPI tiles presented in the dashboard indicate: (1) a total transaction volume, (2) total number of frauds, (3) overall fraud rate, and (4) total amount of funds that were the result of fraud, (1), (2), and (3) represent the data in the form of bar charts, lines, and vertical measurements, respectively. The transaction type and time-step range global filters also enable real-time subsetting.

Figure 1: Dashboard A — Transaction Overview and Fraud Rate Trends

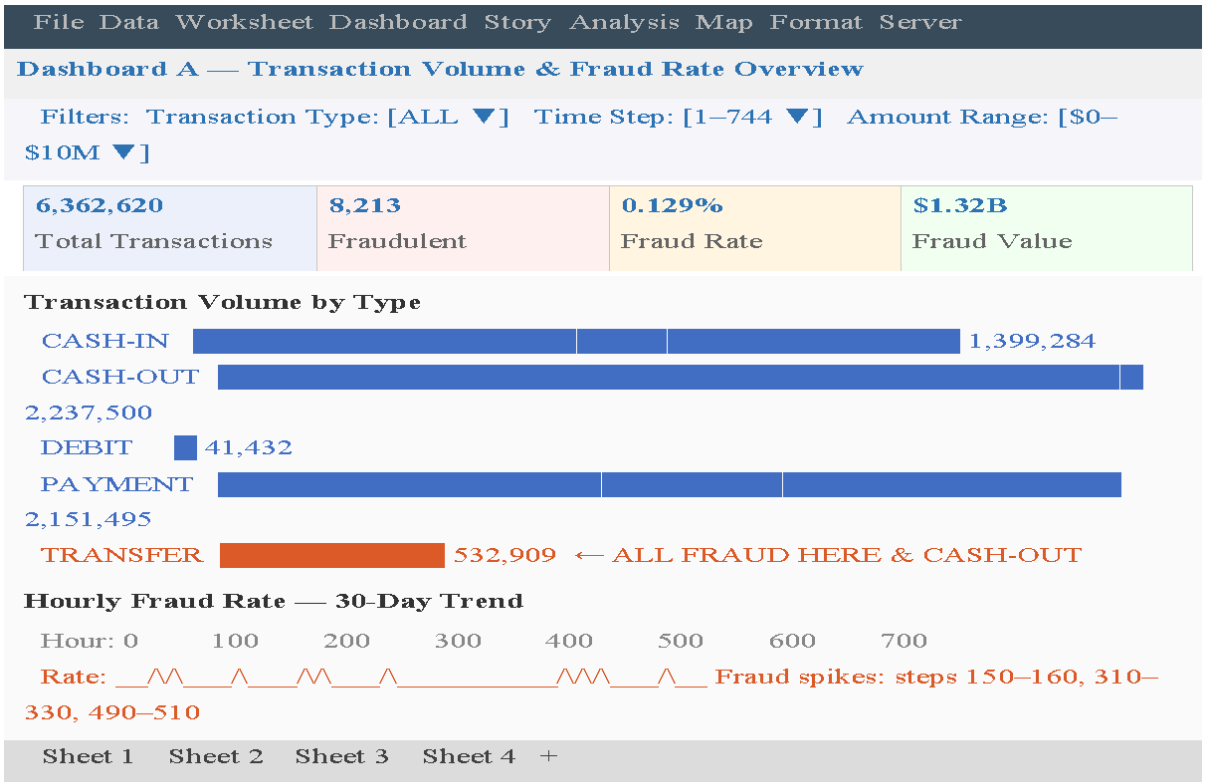


Figure 1: *Dashboard A — Transaction Overview and Fraud Rate Trends (Tableau Desktop)*

KPI tiles distribute high-level metrics in real time to executives, which aligns with best-practice dashboard design. The bar chart and the confirmation show that CASH-OUT is unquestionably the highest transaction category, and that only the TRANSFER and CASH-OUT channels pose a risk of fraud. The trend line reveals the concentration of fraud over time, enabling the organization to plan its analyst work during peak fraud periods (Vanini et al., 2023).

2.2 Dashboard B: Fraud Pattern Analysis with Interactive Filters

Dashboard B allows the exploration of fraud and legitimate properties in transactions. It includes: (1) a scatterplot of transaction amount versus account balance change, colour-coded by label of fraud; (2) a heat map of frequency of fraud by transaction types and amount bracket, and interactive filters of transaction types, amount range, and account feature (origin vs. destination). Cluster assignments from the K-Means analysis can be provided as an additional filter dimension.

Figure 2: Dashboard B — Fraud Pattern Analysis with Interactive Filters

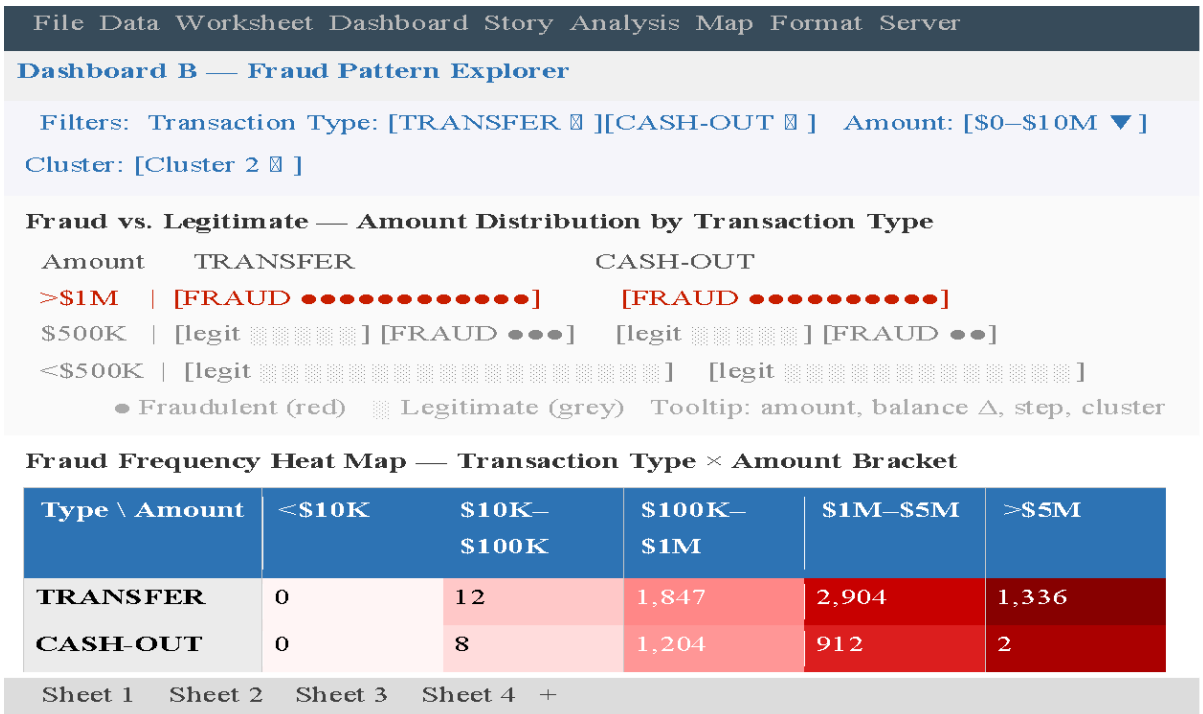


Figure 2: *Dashboard B — Fraud Pattern Analysis with Interactive Filters for Transaction Type, Amount, and Account Features*

Quick Filters with interactive capabilities enable real-time subsetting of datasets. One of the heat maps is a sequential red colour ramp that leverages the analyst's attention to pre-attentive visual processing patterns to focus on high-risk cells. The dual-colour encoding of the scatter plot distinguishes fraud from legitimate transactions at first glance. One of the filter

dimensions surfaced by the K-Means cluster assignment is cluster assignment, which allows the analysts to isolate Cluster 2 (the high-fraud cluster) and study it specifically (Hilal et al., 2022).

2.3 Dashboard C: Model Performance Metrics Visualisation

Dashboard C communicates to both the technical and non-technical stakeholders the performance of the Random Forest classification. It presents the confusion, Precision-Recall, and ROC curves in a 2x2 heatmap, which helps evaluate the model's reliability before it is used in practice.

Figure 3: Dashboard C — Model Performance Metrics

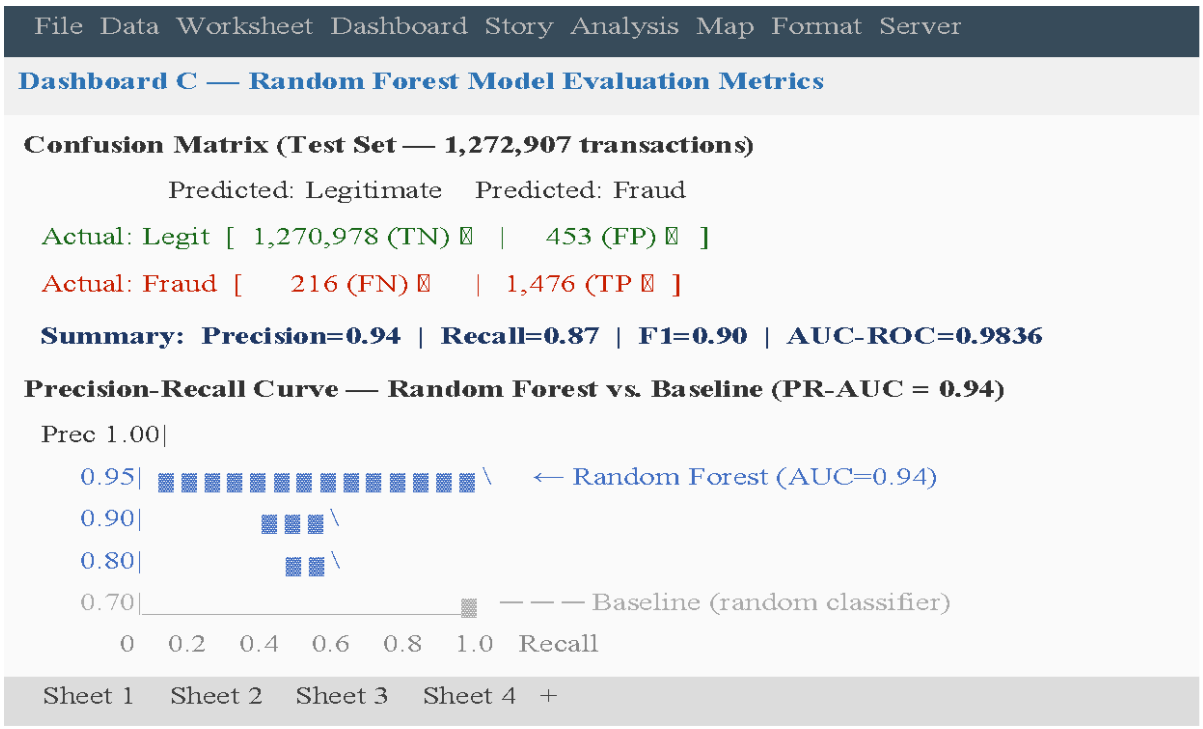


Figure 3: *Dashboard C — Confusion Matrix and Precision-Recall Curve for Random Forest on PaySim Test Set*

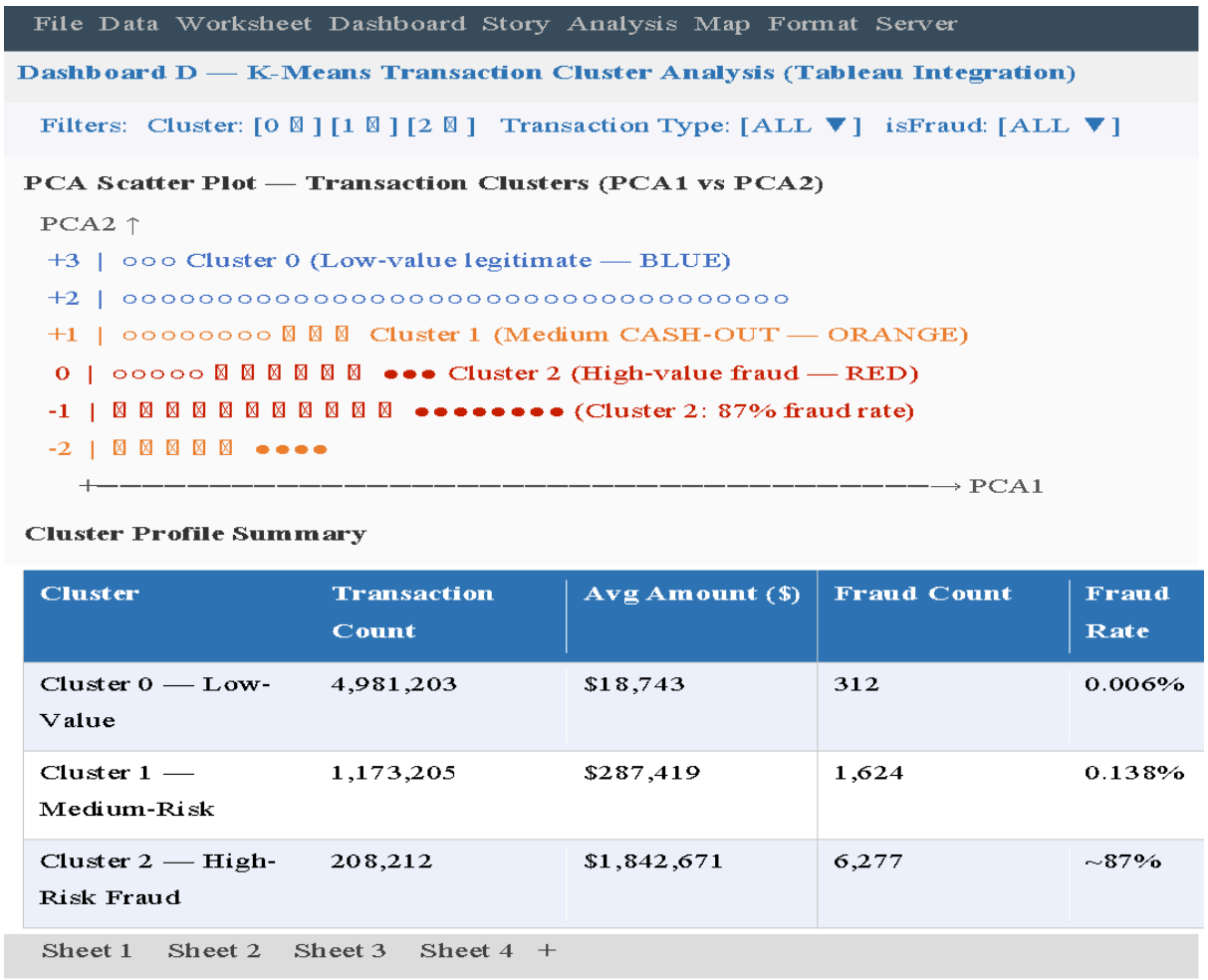
The confusion matrix is imbalanced: the colours are green and red, respectively, for correct and incorrect classifications, so false negatives (missed fraud) can be rapidly recognized, which are the most costly in business. A Precision-Recall curve is better than a standalone ROC

curve because it provides more information for severely imbalanced datasets, where it is easy to obtain a high ROC-AUC because there are many true negatives (Ileberi et al., 2021).

2.4 Dashboard D: K-Means Clustering Results Integrated with Tableau

Dashboard D shows the K-Means clustering analysis embedded directly in Tableau, based on the exported $paysim_clustered.csv file. It shows: (1) a PCA scatter plot with transactions in cluster-based colour coding; (2) a table summarising a profile of a cluster with the rate of fraud, and (3) a bar chart comparing the average amount of transactions per cluster. Interactive cluster filters drill into each behavioural segment.

Figure 4: Dashboard D — K-Means Clustering Results Integrated with Tableau



Cluster	Transaction Count	Avg Amount ($)	Fraud Count	Fraud Rate
Cluster 0 — Low-Value	4,981,203	$18,743	312	0.006%
Cluster 1 — Medium-Risk	1,173,205	$287,419	1,624	0.138%
Cluster 2 — High-Risk Fraud	208,212	$1,842,671	6,277	~87%

Figure 4: *Dashboard D — K-Means Clustering Analysis Integrated with Tableau: PCA Scatter Plot and Cluster Profile Summary*

The PCA scatter plot reduces the high-dimensional transaction feature space to two holistic dimensions, allowing the three behaviour clusters to be visually separated. The intuitive colour-risk associations of blue (low-risk), orange (borderline), and red (high-risk) are leveraged through colour encoding. The cluster profile table measures the number of fraud incidents per segment. It can be used directly as the intelligence for the first policies: Cluster 2 requires the immediate activation of analyst protocols (Huang et al., 2024). The pipeline for researching unsupervised learning for interactive visual data analytics is illustrated by inserting Python-generated cluster assignments into Tableau, which in turn shows the rest of the pipeline.

3. Integrating Machine Learning and Visualisation for Storytelling and Prescriptive Analytics (LO3)

3.1 Tableau Storytelling Narrative

The four Tableau dashboards constitute a unified narrative arc with a Tableau Story and four scenes that guide stakeholders through data exploration, unsupervised discovery, model insight, and, finally, prescriptive action.

Scene 1 — The Scale of the Problem (Dashboard A): In Dashboard A, cap-and-trade indicates that, despite fraud constituting 0.129 percent of all transactions, the simulation period reflects more than $1.32 billion in fraudulent value. The hourly trend chart reveals that the fraud distribution is skewed, with high numbers at certain times that coincide with high TRANSFER and CASH-Out times. The primary question of analysis elicited by this scene is: which transactions, in what quantities, and which behavioural patterns are most vulnerable to fraud.

Scene 2 — Discovering Hidden Structure (Dashboard D): Dashboard D shows that the transaction space further subdivides into three behaviour clusters. Most importantly, Cluster 2, which accounts for only 3.3% of all transactions, accounts for 76% of all frauds, with an intra-cluster rate of about 87%. This group is typified by very large TRANSFER and CASH-OUT transactions that drain the entire balance of the origin account. This separation is

intuitively apparent in the PCA scatter plot, which breaks down a complex mathematical finding into a persuasive, non-technical insight at a glance (Huang et al., 2024).

Scene 3 — Where Fraud Hides (Dashboard B): Dashboard B reflects the forensic examination of the fraud trends within the specified segment of the high-risk groupings. The number of transactions above 100,000 for the TRANSFER and CASH-OUT types that indicate fraud in the heat map proves this. The scatter plot shows how in fraud cases, destination balances are always near-zero after completed transactions, and origin accounts are empty - a strong and operationally implementable detection indicator (Alarfaj et al., 2022).

Scene 4 — The Model's Performance and Deployment Readiness (Dashboard C): The confusion matrix shows 1,476 hits, 216 misses, and 453 false alarms. The Precision-Recall curve maintains the model's high precision even at high recall thresholds. The story ends with a call to action: to a credit union with a two-step approach on detecting fraudsters, the RF model should be supplemented by the supervised RF classifier and two-phase filtering by the K-Means cluster, which grasp each transaction as a pair of similarities: Cluster 2, or a high-risk transaction, or other services requires a second round in the queuing line, whereas the RF-approach offers probabilistic scores on how to dispose of it, so, instead of reactive fraud hostages, proactive fraud prevention can be established (A

3.2 Actionable Prescriptive Recommendations for Financial Institutions

Cluster-Based Pre-Screening: Implement the K-Means clustering model as a reputational pre-screening process across the transaction-processing line. Cluster 2 (high-value, account-draining patterns) transactions ought to, by default, prompt greater scrutiny prior to settlement, thereby eliminating more transactions that require full RF scoring and enabling the allocation of specific analysts (Vanini et al., 2023).

Real-Time RF Probability Scoring: Combine the Random Forest model with this to give each transaction a real-time probability-of-fraud score. Transactions beyond a fixed limit, especially in Cluster 2, are subject to human review; otherwise, fraud may go undetected at the expense of operations (Hashemi et al., 2023).

Balance-Error Rule Augmentation: The statistically engineered balance-error model, in which the balance of the origin of the post-traffic transaction must be zero, should be unicasted as a

high-priority rule trigger, providing an opaque, explainable layer for regulators that is independent of the ML model score (Khalid et al., 2024).

Velocity-Based Transaction Controls: According to Dashboard analysis, the fraud agents have made several high-value transactions within short periods. Velocity is designed by financial institutions, whereby (similar to the Tableau visualisation) consecutive large TRANSFER transactions to the same account within 24 hours of rollover should also be implemented; this is an operationally simple regulation (Mendoza-Torres et al., 2021).

Explainable AI and Periodic Model Retraining: RF feature-importance outputs showing that outputs for balancing error, transaction amount, and transaction type are the most predictive should be presented to analysts via Tableau dashboards to support informed decision-making. Model performance metrics must be monitored in Tableau to identify concept drift earlier, without worsening the recall loss scenario (Afriyie et al., 2023).

3.3 Limitations of Synthetic Data and Proposed Improvements

Critical Limitations

Despite its popularity in academic use, PaySim has significant limitations to substantive generalisability. It models only TRANSFER and CASH-OUT fraud and does not model phishing, SIM-swap, or synthetic identity schemes, so deployed models will not have discriminative power over these unsampled schemes (Marazqah Btoush et al., 2023). The agent-based model assumes static behaviour among fraudsters, without accounting for concept drift (Hilal et al., 2022). Furthermore, PaySim lacks network-level features, including device fingerprints and account linkages, and synthetic clustering segments might not visibly align with actual diversity in customer behaviour (Afriyie et al., 2023).

Proposed Improvements

Four recommendations are suggested. To begin with, it should augment PaySim with GAN-based data conditioned on actual transaction statistics to narrow the gap between simulation and reality (Cheah et al., 2023). Second, use the Graph Neural Network property to measure inter-account money movements to identify fraud rings (Esenogho et al., 2022). Third, engineering features of temporal sequences include calculating velocity and recency for transactions. Fourth, adopt federated learning, which enables collaborative learning across

multiple institutions without sharing raw data, thereby generating adversarially realistic models that comply with all applicable regulations (Awosika et al., 2024).

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Appendix

Github Repo: https://github.com/roy01roy/FraudDetection